A Guide for Software Architects: Common Mistakes and Best Practices

Software architects play an invaluable role in the digital transformation of an organisation. To make a mark, they must imbibe certain qualities and avoid common errors.

Every organisation aims to perform better than its competitors, provide smart services to its customers, and be agile in adopting new business strategies. In this context, software architecture is a differentiator as well as a disruptor. New digital business models and unrelenting focus on client engagement are unlocking new, exciting initiatives and creating an unprecedented demand for software architects who can comfortably bridge any gap between business and technology.

Organisations today realise the importance of software architects and their new role in digital transformation. Gone are the days of extensive documentations, long timelines and rigid technology models. Enterprise architecture delivers great value and cuts costs and complexity by aligning technology tools with strategic business goals.

Modern software architecture has five components: strategy (business transformation), application modernisation (automation, cloud, microservices, APIs), connected intelligence (Big Data, AI/ML, IoT, 5G and edge technologies), trust (blockchain, cybersecurity, data privacy, governance and transparency) and realisation (Agile, DevSecOps and CI/CD). The software architect in this digital age must be well versed in all of these.

It is evident that next generation technologies like AI/ML, generative AI, cloud, Big Data analytics, cybersecurity, and platform as a service are going to change our world in ways we cannot imagine today. Adoption